Assignment 7

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Are subject relatives easier to process than object relatives (shifted log-normal likelihood)?

This is a classic question from the psycholinguistics literature: Are subject relatives easier to process than object relatives? The data come from Experiment 1 in a paper by Grodner and Gibson 2005.[1]

Scientific question: Is there a subject relative advantage in reading?

Grodner and Gibson 2005 investigate an old claim in psycholinguistics that object relative clause (ORC) sentences are more difficult to process than subject relative clause (SRC) sentences. One explanation for this predicted difference is that the distance between the relative clause verb (*sent* in the example below) and the head noun phrase of the relative clause (*reporter* in the example below) is longer in ORC vs. SRC. Examples are shown below. The relative clause is shown in square brackets.

(1a) The *reporter* [who the photographer *sent* to the editor] was hoping for a good story. (ORC)

(1b) The *reporter* [who *sent* the photographer to the editor] was hoping for a good story. (SRC)

The underlying explanation has to do with memory processes: Shorter linguistic dependencies are easier to process due to either reduced interference or decay, or both.

In the Grodner and Gibson data, the dependent measure is reading time at the relative clause verb, (e.g., *sent*) of different sentences with either ORC or SRC. The dependent variable is in milliseconds and was measured in a self-paced reading task. Self-paced reading is a task where subjects read a sentence or a short text word-by-word or phrase-by-phrase, pressing a button to get each word or phrase displayed; the preceding word disappears every time the button is pressed.

For this experiment, we are expecting longer reading times at the relative clause verbs of ORC sentences in comparison to the relative clause verb of SRC sentences.

```r
load("df_gg05_rc.rda")
head(df_gg05_rc)
```

```
## # A tibble: 6 x 7
##    subj  item condition    RT residRT qcorrect experiment
##   <int> <int> <chr>     <int>   <dbl>    <int> <chr>
## 1     1     1 objgap      320  -21.4         0 tedrg3
## 2     1     2 subjgap     424   74.7         1 tedrg2
## 3     1     3 objgap      309  -40.3         0 tedrg3
## 4     1     4 subjgap     274  -91.2         1 tedrg2
## 5     1     5 objgap      333   -8.39        1 tedrg3
## 6     1     6 subjgap     266  -87.3         1 tedrg2
```

You should use a sum coding for the predictors. Here, object relative clauses (`"objgaps"`) are coded $+1/2$, subject relative clauses $-1/2$.

[1] Grodner, D., & Gibson, E. (2005). Consequences of the serial nature of linguistic input for sentenial complexity. Cognitive Science, 29(2), 261-290.

```r
df_gg05_rc <- df_gg05_rc %>%
  mutate(c_cond = if_else(condition == "objgap", 1/2, -1/2))
```

Before fitting, also clean the reading times: exclude relative clause verb reading times below 150 ms, since responses that fast probably are an anticipatory button-press instead of actually reading the target word and reacting. Report how many trials this excludes.

```r
n_before <- nrow(df_gg05_rc)
df_gg05_rc <- df_gg05_rc %>% filter(RT >= 150)
n_before - nrow(df_gg05_rc)
```

```
## [1] 4
```

Reading times can never be shorter than some minimum time (the time needed for basic perceptual other processes necessary for reading). We can capture this by shifting the log-normal distribution away from zero by an additional parameter (`ndt`, non-decision time). You can now fit a "maximal" model (correlated varying intercept and slopes for subjects and for items) assuming a **shifted log-normal likelihood**.

(a) Specify the statistical model and define priors for all parameters in the model, including the additional shift parameter `ndt`. *Hint:* To get a feel for reasonable ranges of values for the μ and σ parameters of a (shifted) log-normal distribution, you might want to simulate log-normal data for different values using `rlnorm()` and then add a constant shift to it. Also think about what constraints `ndt` needs to satisfy relative to the observed data for the model to be identifiable, and where the bound for its prior should come from. Should it be extracted from this particular sample, or fixed in advance?

Caution: `brms` uses a `log` link for `ndt` by default (check `?brms::shifted_lognormal`), which means that, depending on how you specify the model, the units your prior and posterior for `ndt` are expressed in may *not* be milliseconds. Before interpreting or setting a prior for `ndt`, check `make_stancode()` for your model to see exactly how `ndt` is parametrized.

(b) Examine the effect of relative clause attachment site (the predictor `c_cond`) on reading times `RT` (β_1).

(c) Estimate the median difference between relative clause attachment sites in milliseconds, and report the mean and 95% CI. Remember to take the estimated shift `ndt` into account when you transform back to the millisecond scale.

(d) Inspect individual differences in the subject relative advantage effect. Are there participants with effects in opposite directions?

(e) Do a sensitivity analysis. What is the estimate of the effect (β_1) under different priors? What is the difference in milliseconds between conditions under different priors? Does the choice of prior on `ndt` also affect the results?

(f) So far `ndt` has been a single constant shared by everyone. In reality, people probably differ in their non-decision time too. Extend the model so that `ndt` varies by subject (`ndt ~ 1 + (1 | subj)`), and think carefully about what prior to put on it. Keep in mind the `log` link caution from part (a). *Hint:* a prior with mean around 5.5 to 6.5 on the *linear predictor* scale for `ndt`'s intercept, together with a weakly-informative prior on its between-subject SD, could work out well here. Think about why, and what those numbers correspond to once exponentiated back to milliseconds. You also need to to give the sampler some sensible starting values.